

Assignment – I

Part - I

Binary Systems, Boolean Algebra & Logic Gates

1. What are the characteristics of Digital Systems.
2. Convert (15)₁₀ to Binary
3. Explain about Diminished Radix complement
4. What is meant by parity bit?
5. Define duality property.
6. Define binary logic?
7. Which gates are called as the universal gates? What are its advantages?
8. State the associative property of Boolean algebra.
- 9 Explain classification of Number system
10. What is a prime implicate?

Part -II

- 1 a) Convert the following numbers
i)(163.789)₁₀ to Octal number ii)(11001101.0101)₂ to base-8 and base-4
iii)(4567)₁₀ to base2 iv) (4D.56)₁₆ to Binary
b) Subtract (111001)₂ from (101011)₂.
2. a) Represent the decimal number 3452 in i)BCD ii)Excess-3
b) perform (-50)-(-10) in binary.
c) Determine the value of base x if(211)_x=(152)₈
3. a) Convert the following numbers
i) (250.5)₁₀ = ()₂ ii) (673.23)₁₀ = ()₈ iii)(101110.01)₂=()₈
b) Convert the following to binary and then to gray code (AB33)₁₆
c) Perform the following Using BCD arithmetic (7129)₁₀ + (7711)₁₀
4. Explain the Binary codes with examples?
5. a) What is Digital System? Characteristics of digital systems.
b) Explain the difference between analog and digital systems
6. a) Design the circuit by Using NAND gates $F = ABC' + DE + AB'D'$
b) Simplify and implementation the following SOP function using NOR gates
 $F(A,B,C,D) = \sum m(0,1,4,5,10,11,14,15)$
7. Convert the following
a)(1AD)₁₆=()₁₀ b) (453)₈=()₁₀ c) (10110011)₂=()₁₀ d) (5436)₁₀=()₃.
- 8 a)Explain binary to Gray & Gray to binary conversion with example?
b) State and Explain the DeMorgan's Theorem and Consensus Theorem
- 9) Convert the following numbers
i) (615)₁₀ = ()₁₆ ii) (214)₁₀ = ()₈ iii)(0.8125)₁₀=()₂
iv) (658.825)₁₀=()₈ v)(54)₁₀=()₂
- 10) Explain the Excess-3 code? Write about Error correction & Detection?

Part -III

1. Simplify the Boolean expression using K-MAP
 $F(A,B,C,D,E) = \sum m(0,1,4,5,16,17,21,25,29)$
2. Simplify the Boolean expression using K-MAP
 $F(A,B,C,D) = \sum m(1,2,3,8,9,10,11,14) + d(7,15)$
3. Simplify the Boolean expression using K-map and implement using NAND gates
 $F(A,B,C,D) = \sum m(0,2,3,8,10,11,12,14)$
4. Simplify the Boolean expressions to minimum number of literals
 - i) $(A + B)(A + C')(B' + C')$
 - ii) $AB + (AC)' + AB'C (AB + C)$
 - iii) $(A+B)' (A'+B)'$
5. Reduce the expression $f(x,y,z,w) = \pi M(0,2,7,8,9,10,11,15) .d (3,4)$ using K-Map?
6. Simplify the Boolean expression using K-map?
 $F(A,B,C,D,E) = \sum m(0,2,4,6,9,11,13,15,17,21,25,27,29,31)$
7. Obtain the a) SOP b) POS expression for the function given below
 $F(A,B,C,D) = \sum m(0,1,2,5,8,9,10)$
8. a) Simplify the Boolean expressions to minimum number of literals
 - i) $X' + XY + XZ' + XYZ'$
 - ii) $(X+Y)(X+Y')$
9. b) Obtain the Complement of Boolean Expression
 - i) $A+B+A'B'C$
 - ii) $AB + A(B+C) + B'(B+D)$
10. Determine the minimal sum of product form of
 - a) $f(w,x,y,z) = \sum m(4,5,7,12,14,15) + d(3,8,10)$
 - b) $F(A,B,C,D) = \pi M(0,3,5,6,8,12,15)$

Part-IV

1. Define Half adder and full adder
2. Define subtractor?
3. Explain The Half adder? Implement the full adder using two half adders.
4. Explain The Half subtractor? Implement the full adder using two half subtractor.